

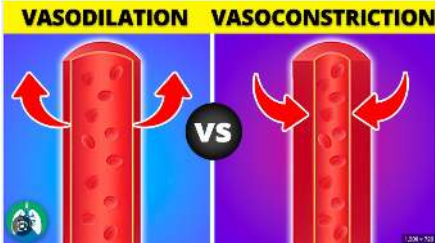
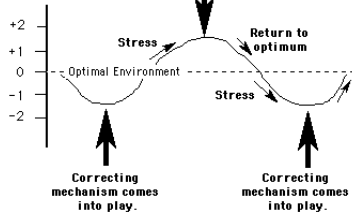
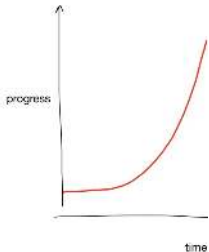
Biology Syllabus

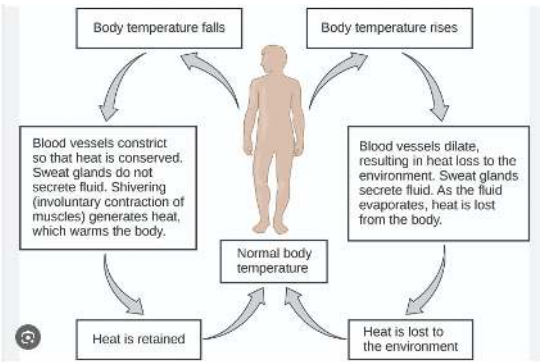
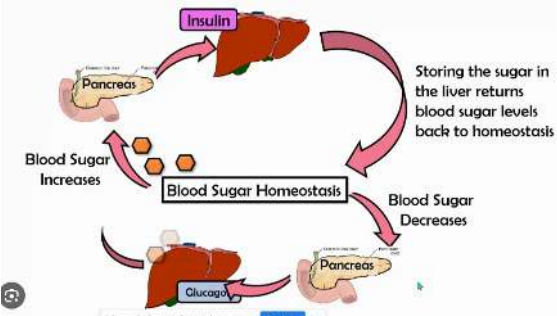
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Module 8: Non-infectious Disease and Disorders

Homeostasis

Inquiry question: How is an organism's internal environment maintained in response to a changing external environment?

Syllabus dot point	Summaries
- construct and interpret negative feedback loops that show homeostasis by using a range of sources, including but not limited to: (ACSBL101, ACSBL110, ACSBL111) - temperature (ACSBL098) - glucose	Homeostasis : the maintenance by an organism of a relatively constant internal state, regardless of external changes in the environment Homo = same, stasis = state - composed of two main stages: 1. Detecting change - stimulus and receptors 2. Counteracting the change - response and effector → cool-down - sweat - vasodilation (turn red) → warm up - shiver - vasoconstriction - goosebumps  → feedback loop stimulus ↓ sensor ↓ control ↓ effector - negative = stops E.g. glucose & body temperature  - positive = increases E.g. oxytocin (contractions until baby comes out)  - body temperature

	 - glucose 			
● investigate the various mechanisms used by organisms to maintain their internal environment within tolerance limits, including: - trends and patterns in behavioural, structural and physiological adaptations in endotherms that assist in maintaining homeostasis (ACSBL099, ACSBL114)	Adaptations: <table><tr><td> Structural : physical characteristics of an organism to enable its survival </td><td> Behavioural : the way an organism acts to enable its survival (whole body) </td><td> Physiological : the way the organism's body functions to enable its survival (part of the body) </td></tr></table> → Water balance - structural: Koalas have a large cecum, a pouch in their intestine where fermentation of eucalyptus leaves takes place. This cecum helps extract every last bit of moisture and nutrients from their tough diet. - behavioural: Emus are most active in the cooler mornings and evenings. This reduces exposure to the hottest part of the day, minimizing water loss through sweating and respiration. - physiological: Australian sea lion have metabolic water production: When breaking down fats and proteins for energy, sea lions produce metabolic water as a byproduct. This internal water source helps to offset water loss through breathing and sweating. → temperature regulation - structural: Koalas have a thick coat of fur that provides some insulation. This fur helps to trap a layer of air around their bodies, which can act as a barrier to both heat and cold. - behavioural: Emus are adept at finding and utilizing shade during the hottest parts of the day. - physiological: Sea lions have a thick layer of blubber acts as a natural wetsuit. This blubber insulates their bodies in cold water, preventing heat loss.	Structural : physical characteristics of an organism to enable its survival	Behavioural : the way an organism acts to enable its survival (whole body)	Physiological : the way the organism's body functions to enable its survival (part of the body)
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- internal coordination systems that allow homeostasis to be maintained, including hormones and neural pathways (ACSBL112, ACSBL113, ACSBL114)

→ hormones

- the endocrine system assist in the regulation of homeostasis

- main components is hormones

: chemical messengers secreted by endocrine glands

- these hormones are transported through the bloodstream

- causes cells to change their activity that will enable them to maintain homeostasis

- hormones causes these cells (target cells) to change by influencing the activity of particular enzymes alongside the concentration of these enzymes

- hormones include

- Melatonin : hormone brain produces in response to darkness
- Insulin : hormone produced to maintain blood glucose levels
- Glucagon : hormone that promotes the breakdown of glycogen to glucose

- pituitary gland

- Works simultaneously with hypothalamus (area of the brain that produces hormones that control body temp, heart rate, hunger, etc.)
- Secretes antidiuretic hormone (ADH) that regulates the concentration of water within body
- If body water is too low, hypothalamus stimulates pituitary gland to release ADH to conserve water

- pancreas:

- Chemoreceptors (detect high levels of glucose) stimulate the production of insulin = negative feedback loop

→ neural pathways

• receptors

: detect stimuli in the form of changes from a set point outside tolerance limits

- Intereceptors (within body)
- Thermoreceptors (temperature)
- Chemoreceptors (chemical concentration)

- central nervous system composes of the brain and spinal cord and communicates with body through electrochemical impulses

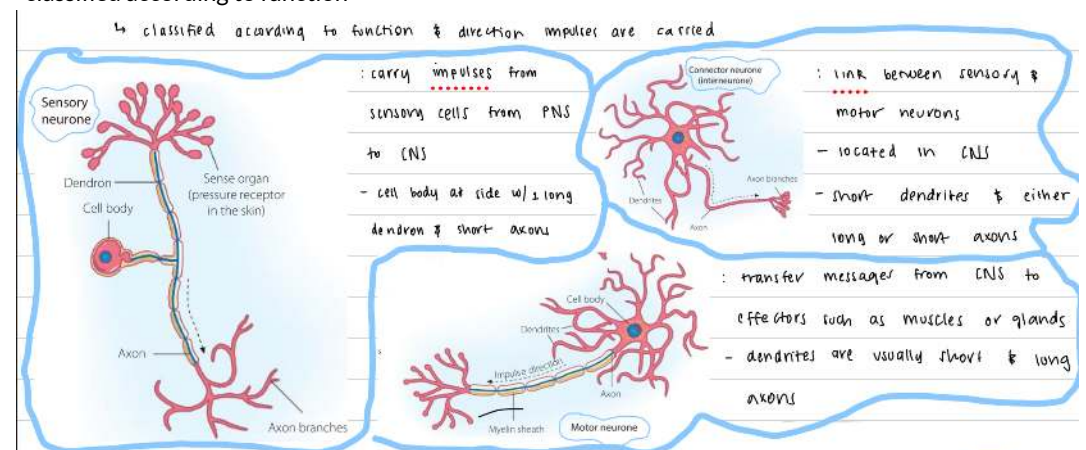
• neurons

: nerve cells that transmit messages

- all different but contain three common structures

1. Cell body - nucleus & other organelles
2. Dendrites - extension of the cytoplasm that receive the impulse (messages) & conduct these to cell body
3. Axon - extension of the cytoplasm that conduct message away from cell body

- classified according to function



- mechanisms in plants that allow water balance to be maintained (ACSBL115)

Transpiration

: the evaporation of water from the stomata of leaves by lifting water and dissolved ions from the roots and up the stems to the top of plants

- allowing for the regulation of temperature

	- to maintain water balance in plants: - Shiny waxy cuticles = to ensure epidermal cells are waterproof - orient leaves away from sunlight - reduce leaf size - flower size - shed leaves - Stomata opening = occurs during cold periods of the day - Hairy leaves & rolled leaves = trap water in stomata for humid air to circulate
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Causes and Effects

Inquiry question: Do non-infectious diseases cause more deaths than infectious diseases?

Syllabus dot point	Summaries
• investigate the causes and effects of non-infectious diseases in humans, including but not limited to: - genetic diseases	Non-infectious : not caused by pathogens → genetic disease : genetic abnormalities E.g. sickle cell anemia, haemophilia & down syndrome Case study: sickle cell anaemia • cause & effects - GAG = glutamic acid → GTG = valine - Sickle red blood cells • symptoms - Blood clots - Less transportation of oxygen • treatment - Short: blood transfusion - Medium: bone marrow transplant - Long: gene therapy e.g. CRISPR • prevention - Gene counselling = won't be passed down - Other options e.g. adoption
- diseases caused by environmental exposure	→ environmental disease : materials in the environment that can cause diseases - long exposure to toxic substances, pollutants, radiation, etc. Case study: Minamata disease (mercury poisoning) • cause & effects - Mercury build-up - Mercury consumption • symptoms - Convulsions - Crazy mental state - Permanent coma • treatment - Chelating agents: remove mercury from the blood • prevention - Avoiding mercury-filled food
- nutritional diseases	→ nutritional disease : excessive or insufficient food intake Case study: scurvy (vitamin C deficiency) • cause & effects - Insufficient intake of vitamin C • symptoms - Bone weakness - Spontaneous bleeding - swelling • treatment - Consume adequate vitamins

	• prevention - Consume adequate vitamins															
- cancer	→ cancer : a disease in the cells of the body when abnormal cells divide in an uncontrolled way - types of tumours - Benign → cells do not spread past tumour (not classified as cancer but can develop into it) - Malignant → abnormal cancerous cells and their cells replicate more quickly and in a more disorganised manner than normal cells, redirect the nutrients to themselves and away from adjacent normal tissue, and can invade surrounding tissue Case study: breast cancer • cause & effects - It can be genetic, environmental/exposure or nutritional; however, generally: Cells gone wild = No Function or Fast reproducing - Genetic: genetic coding causes an increase in susceptibility to cancer, such as breast cancer. - Environmental/exposure: Smoking causes lung cancer - Nutritional: a lot of spicy food can cause bowel cancer • symptoms - Lump - Change in breast size, shape or appearance - Dimpling, redness, pitting or other changes in the skin - Change in nipple appearance or skin surrounding the nipple (areola) - Abnormal or bloody fluid from the nipple • treatment - Chemotherapy: Chemical treatment that kills all fast reproducing cells such as hair, digestive cells, cancer - Radiation therapy: the patient lies down and gets a 'tattoo', and the radiation targets that area, killing all the cells - Surgery: total movement of the affected area • prevention - Gene counselling - Avoiding carcinogenic substances															
• collect and represent data to show the incidence, prevalence and mortality rates of non-infectious diseases, for example: - nutritional diseases - diseases caused by environmental exposure	<table><tr><th></th><th>Nutritional disease</th><th>Environmental exposure</th></tr><tr><th>Disease</th><td>Type 2 Diabetes</td><td>Melanoma</td></tr><tr><th>Description</th><td>: a condition where the body can become insulin resistance and can no longer produce it. - this leads to a buildup of glucose in the blood - causes include genetic factors, sedentary lifestyles and poor diet </td><td>: the most lethal type of skin cancer, developing from mutation of melanocytes - caused by exposure to ultraviolet radiation, usually from the sun as a result of lifestyle factors such as levels of outdoor activity or genetic factors such as melanin levels </td></tr><tr><th>Incidence</th><td>- people with Type 2 Diabetes has quadrupled in the last 30 years. Predicted increase in adults with diabetes between 2010 and 2030 (20% in developed countries, 69% in developing countries)</td><td>- worldwide there were 324,635 new cases of melanoma in 2020. - In 2013, there were 12 744 new cases of melanoma diagnosed in Australia - In Australia, there were an estimated 17,756 new cases diagnosed in 2020 (11% of all new cases of cancer diagnosed in 2020) </td></tr><tr><th>Prevalence</th><td>- 1 in 11 adults aged 20 – 79 years old had diabetes in 2015 (415 million people) 90% of those people had type two</td><td>- worldwide, melanoma is the 17th most common cancer with the highest rates reported in Australia. - In Australia, an estimated 51,697 </td></tr></table>		Nutritional disease	Environmental exposure	Disease	Type 2 Diabetes	Melanoma	Description	: a condition where the body can become insulin resistance and can no longer produce it. - this leads to a buildup of glucose in the blood - causes include genetic factors, sedentary lifestyles and poor diet	: the most lethal type of skin cancer, developing from mutation of melanocytes - caused by exposure to ultraviolet radiation, usually from the sun as a result of lifestyle factors such as levels of outdoor activity or genetic factors such as melanin levels	Incidence	- people with Type 2 Diabetes has quadrupled in the last 30 years. Predicted increase in adults with diabetes between 2010 and 2030 (20% in developed countries, 69% in developing countries)	- worldwide there were 324,635 new cases of melanoma in 2020. - In 2013, there were 12 744 new cases of melanoma diagnosed in Australia - In Australia, there were an estimated 17,756 new cases diagnosed in 2020 (11% of all new cases of cancer diagnosed in 2020)	Prevalence	- 1 in 11 adults aged 20 – 79 years old had diabetes in 2015 (415 million people) 90% of those people had type two	- worldwide, melanoma is the 17th most common cancer with the highest rates reported in Australia. - In Australia, an estimated 51,697
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		- this is expected to rise to 642 million by 2040 - Asia is the epicentre of the epidemic with China and India being the top two countries for type 2 diabetes prevalence	people were living with melanoma at the end of 2012
	Mortality Rate	- currently causes 5 million deaths per year → mostly as a result of cardiovascular disease - type 2 diabetes is expected to become the seventh most prevalent cause of death globally by 2030.	- worldwide there were 57,043 melanoma-related deaths - In Australia, 1,408 deaths occurred in 2022 (2.6% of cancerous deaths)

Epidemiology

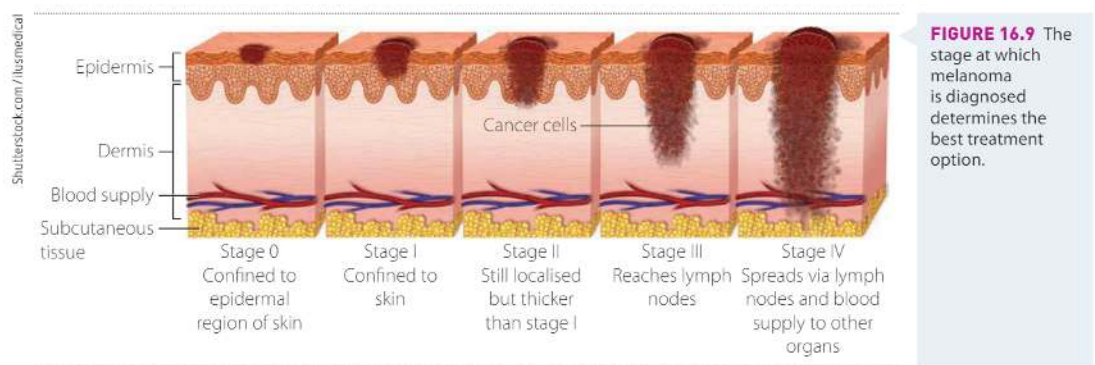
Inquiry question: Why are epidemiological studies used?

Syllabus dot point	Summaries								
● analyse patterns of non-infectious diseases in populations, including their incidence and prevalence, including but not limited to: - nutritional diseases - diseases caused by environmental exposure	Epidemiology : the study of patterns of disease in populations. - An epidemiological study can: - determine the cause of a disease and which populations are affected by the disease - help to develop strategies to control disease and improve public health - evaluate the effectiveness of strategies in place to treat/control disease. - analyse data to identify patterns and trends amongst the population including incidence, prevalence and mortality rates - identify areas and ways in which to improve the overall health of the population Incidence : the number of new cases reported in a specific time period Prevalence : the proportion of the population that have the disease at a particular point in time (used to calculate survival rate) Mortality rates : the number of deaths due to a particular disease in a specific time period Age-standardised rate : a measure of what the rate would be if the population had a standard age structure → nutritional disease : diabetes (type 2) <table><tr><th>Incidence</th><th>Prevalence</th><th>Mortality Rates</th><th>Analysis</th></tr><tr><td>- increased 60% since 2000</td><td>- 85% of diabetes are type 2 - grew in prevalence after the age of 45</td><td>- Australia's 7th leading cause of death in 2020 (men)</td><td>- high rates indicate prevalence is due to poor diet and exercise - older people are most susceptible to diabetes due to decreased insulin production</td></tr></table> → environmental exposure : melanoma	Incidence	Prevalence	Mortality Rates	Analysis	- increased 60% since 2000	- 85% of diabetes are type 2 - grew in prevalence after the age of 45	- Australia's 7th leading cause of death in 2020 (men)	- high rates indicate prevalence is due to poor diet and exercise - older people are most susceptible to diabetes due to decreased insulin production
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Incidence	Prevalence	Mortality Rates	Analysis
- In 2013, there were 12 744 new cases of melanoma diagnosed in Australia - ↑incidence = ↑age - rising due to the aging population	- in 2008 - 2012, 51,697 were still living after being diagnosed	- In 2014, melanoma was the 10th leading cause of cancer deaths in Australia	- younger age group rates have fallen due to government health initiatives e.g. "slip, slop, slap" - greater prevalence due to Australia's location (near the ozone layer = stronger UV rays)

- investigate the treatment/management, and possible future directions for further research, of a non-infectious disease using an example from one of the non-infectious diseases categories listed above

Melanoma
 : the most lethal type of skin cancer, developing from mutation of melanocytes, caused by excessive exposure to UV rays
 - treatment is dependent on the stage of the disease



Treatment	Description
Surgery	: removing the tumour and the skin surrounding it to ensure no cancerous cells are left behind (early stage)
Radiation	: involves high-powered energy beams, such as X-rays and protons, to damage the DNA of the cancerous cells and kill them
Chemotherapy	: intake of drugs to slow the growth of cancerous cells (not found particularly helpful in treating melanoma)
Targeted therapies	: intake of drugs to interrupt the pathway of uncontrolled division of melanoma cells effectively stops the growth and spread of the tumour to other organs
Immunotherapy	: causes the body's own immune system to fight the melanoma (two approaches) 1. use drugs called 'checkpoint inhibitors' as cancerous cells cause the immune system to ignore them but these drugs reverse the effect causing the immune system to recognise and doctor melanoma cells as cancer cells 2. Vaccinations as a method of treatment rather than prevention as an antigen is produced using melanoma cells to allow the immune system to identify and destroy cancerous cells easily

- future directions
 : melanoma

Research	Description
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Immunotherapy	- currently, the use of immunotherapy is not all successful - as cancerous cells have numerous mutations, scientists are still researching ways to combat & interrupt more pathways
Educational programs	- increasing awareness of unsafe sun behaviour - protecting skin = reducing melanoma cases
Genetic investigations	- certain individuals are more susceptible due to genetics - DNA sequencing to identify familial melanoma and reduce risks

- types of epidemiological studies
[not a syllabus dotpoint]

Epidemiological studies

- should:
 - be conducted over a long period of time
 - have a large sample size (thousands +)
 - collect a range of relevant data from both control and test groups (includes data such as age, occupation, etc.)
 - includes a range of participants from a wider population
 - control and test groups
 - collect data on incidence, prevalence, mortality and morbidity rates
 - statistically analyse data to identify patterns and trends regarding the occurrence of the disease
 - identify possible causes of the disease
 - develop a management plan to control or eliminate disease
 - evaluate the effectiveness of control and treatment programs



→ observational studies

- descriptive studies
 - : the first type of study conducted when investigating a disease
 - provide information about the patterns, frequency, area of the population (age, gender, occupation, socioeconomic status, etc), geographical location, and if a particular time period when individuals were affected
 - data collected includes: age, sex, diet, occupation, drinking habits, location of work & home, commonalities
- analytical studies
 - : [after descriptive study is complete] used to collect more data which is statistically analysed to test hypotheses as to the likely cause(s) of the disease
 - the morbidity (number of cases of the disease) and the mortality (percentage of the population that has died from the disease) are used as indicators alongside the incidence and prevalence
 - two types of analytical studies:

- ↳ Case-control studies
 - : compares controls (people without the disease) and the cases (people with the disease) to look at the differences in exposure
 - data is collected such as age, sex, diet, location, lifestyle, occupation and exercise habits and analysed to determine the likely cause
 - E.g. Richard Doll set up a case-control study in London in 1947 comparing patients with lung cancer to patients with other conditions. He was the first to suggest a link between smoking and lung cancer as the results showed most individuals with lung cancer were smokers.
- ↳ Cohort studies

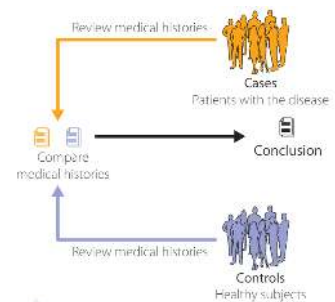


FIGURE 16.11 The principles of a case-control study.



	: studying two or more similar (in terms of e.g. lifestyle choices) groups who are without the disease and analysing the exposure - one group is exposed to possible causes and the other group is not - groups are followed up over an extended period of time to compare the incidence of the disease E.g. following up on the 1947 case-control study, A. B. Hill conducted a cohort study in England in 1951, gathering 40,000 doctors over a 10-year period. One group of doctors (test group) were smokers while the other were non-smokers (control group). It was revealed the test group had a much higher incidence than the control group and also revealed that the greater the number of cigarettes smoked daily, the greater the chance of developing lung cancer → intervention studies : used to test the effectiveness of a treatment or the effectiveness of a public health campaign - the aim is to change the behaviour of a population as a whole to reduce the incidence of a disease • experimental study : a type of intervention study often used to test the effectiveness of a new type of drug - participants are randomised into two groups, one group receiving the trial drug whilst the other receives a placebo (a treatment that appears real but is designed to have no therapeutic effect) - the effects of the 'medication' are recorded and statistically analysed to determine the effectiveness of the drug - however, if a randomised trial is impossible another study is carried out, a quasi-experimental study • quasi-experimental study : the researcher chooses the subjects who receive the drug or treatment rather than it being randomised → errors • random errors : unpredictable variations in data which create an inconsistent effect on measurement - ensuring a large sample size can reduce random error • systematic errors : bias encompassing a constant deviation from what the true value should be ↳ selection bias : bias in selecting subjects to include in the study - influenced by several factors including - Sampling bias - subjects are chosen in a way that does not represent the wider population - Volunteer bias - in which those who volunteered have a vested interest / may be at a higher risk than those who did not volunteer - Healthy worker bias - [in a general sense] those who are in employment are generally healthier than those who are not (will affect data if e.g. exposure is at the place of employment) - Prevalence/incidence bias - only current cases are included ↳ Information bias : errors in taking measurements or recording information - types are - Misclassification bias - some subjects are already suffered or undiagnosed - Recall bias - those affected have a much greater recall of factors than those who are not - Ascertainment bias - all members of the group are not followed up to the same degree - Interviewer bias - The interviewer indirectly leads the participant to the answer - Measurement bias - measurements are overstated or understated consistently - Loss to follow-up bias - not all subjects who began are available at the end of the study
• evaluate the method used in an example of an epidemiological study	Evaluate : make a judgement - criteria

	- Large sample size (1000+) - A long period of study (10+ years) - Use of scientifically approved methods - Collecting ranges of data and analysing results - Reliability: consistent results, compared to other studies, peer-reviewed, control variables, accurate forms of measure, trained professionals (if needed) E.g. Pima Indian Epidemiological study - cohort study in the Pima Indian population - examined the role of physical activity in the development of type 2 diabetes - the study was carried out between 1987 - 2000 (13 years) - 1,728 non-diabetics aged 15-59 years - same individuals are examined every two years and a glucose test & physical examination (height, weight, medical history, etc) test is performed - leisure and occupational activity recorded (inclusive of how many hours sitting, activity to and from work, frequency and duration of each activity) - result: in most age groups, and for males and females, the incidence rate of diabetes was lower in those who had higher levels of physical activity than in those who had low levels of physical activity → evaluation of the above study: - study size of 1798 & 13 year period satisfies the requirements - cohort study (two or more similar groups) is satisfied as non-diabetics were studied from the same areas & age range - consistent testings & follow-up that reduce the likelihood of measurement bias - trained professionals used a valid questionnaire that reduced interviewer and measurement bias - scientifically tested models used to analyse results and determine incidence rates of diabetes related to activity levels - the written report of the study was peer-reviewed before publication - judgement: a valid epidemiological study
● evaluate, using examples, the benefits of engaging in an epidemiological study	Benefits: - benchmark prevalence of disease - monitors changes over time - creates specificity for public health & educational campaigns - identifies patterns - potentially saves lives as it allows the cause of disease to be determined - evaluates effectiveness of treatment to identify if: - Health services are required - Development (time and money) - Further research

Prevention

Inquiry question: How can non-infectious diseases be prevented?

Syllabus dot point	Summaries
● use secondary sources to evaluate the effectiveness of current disease-prevention methods and develop strategies for the prevention of non-infectious disease, including but not limited to: - educational programs and campaigns - genetic engineering	→ educational programs and campaigns : provide information and educate population about effects and risk factors of a disease - successful campaigns have: - An evidence base - Limited number of high priority interventions - Effective performance management - Partnerships between public and private sectors - Communication of accurate information - Political commitment - common campaigns: 'slip, slop, slap, seek, slide' or 'smoking kills' - e.g. QUIT campaign - Smoking is the leading preventable cause in Australia - 15, 500 smoking related deaths per year - Many epidemiological studies have exhibited a clear link between cigarette smoking and reduce life expectancy

- Results show that *"the risk of dying from lung cancer is more than 23 times higher in men and about 13 times higher in women than nonsmokers"* (according to National Institute of Health)
- The QUIT campaign has many strategies including
 - Use of slogans e.g. 'quit for life', 'quit B fit', 'iCanQuit'
 - Graphic images in media & cigarette packaging
 - National helpline
 - Legislation requiring cigarette packaging are out of sign for consumers
 - Legislation requiring strict advertising of tobacco products
 - Increasing pricing on tobacco products
 - Raised legal age of smoking to 21

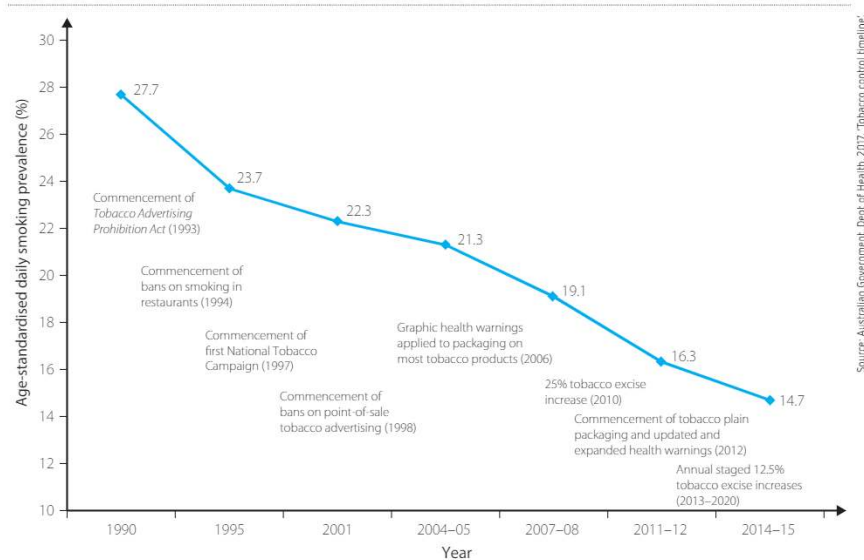


FIGURE 17.8 Age-standardised daily smoking prevalence in the general population of people aged 18 years and older and key tobacco control measures implemented in Australia since 1990

→ genetic engineering

- used to prevent genetic disease through pre-implantation genetic testing (PGT)
- used to prevent nutritional disease such as vitamin A deficiency and combat through golden rice
- e.g. genetic disease: preimplantation genetic testing for deafness:
 - Eggs are harvested, collected & fertilised (through in-vitro fertilisation)
 - Each embryo is tested for mutated gene of deafness
 - Embryo free of gene or carriers were retained, embryo with gene were destroyed

Advantages

- considered more ethical than testing (and then aborting foetus)
- cost saving
- carriers can be identified
- high accuracy

Disadvantages

- ethically against IVF
- discarded embryos is unethical

- e.g. nutritional disease: vitamin A deficiency (VAD)

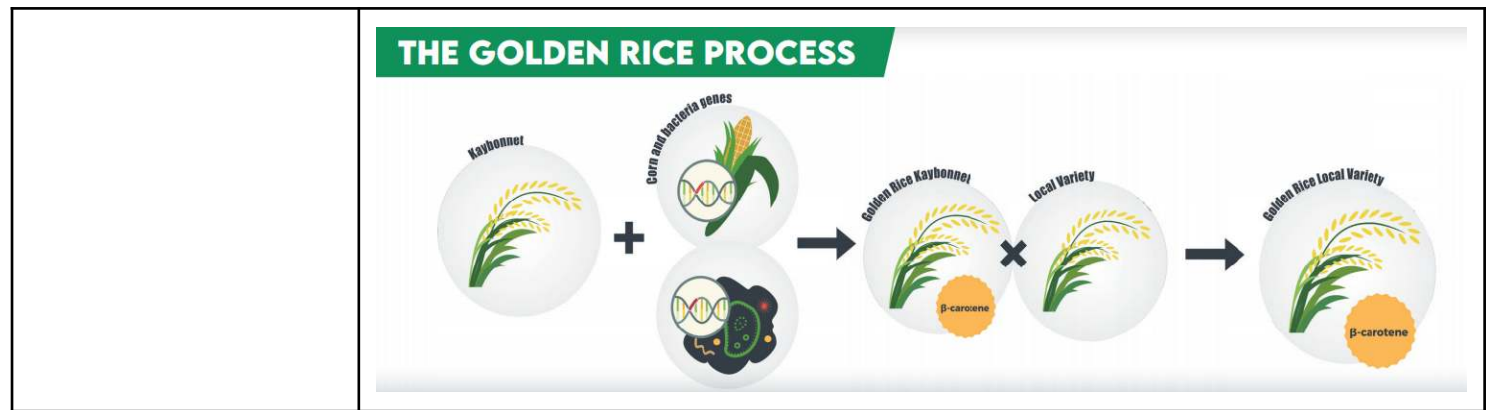
- Golden rice is genetically modified to contain genes from soil bacterium that produces beta-carotene that produces vitamin A
- VAD may lead to blindness and compromised immune system

Advantages

- developing countries have access to nutrition
- rice is a staple food = easy to incorporate
- seeds are easily replantable

Disadvantages

- costly (due to third world countries)
- smaller yield than average rice
- ethical considerations towards genetically modified organisms

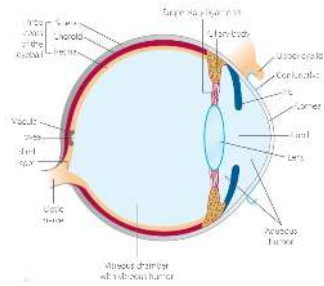


Technologies and Disorders

Inquiry question:

Syllabus dot point	Summaries
- explain a range of causes of disorders by investigating the structures and functions of the relevant organs, for example: - hearing loss	→ Ear : a sense organ that provides a major communication pathway between the external environment and the body - two distinct functions: hearing and balance - anatomy has three sections - Outer ear: pinna & ear canal ending at tympanic membrane (ear drum) - Middle ear: air filled chamber with three small bones (hammer, anvil & stirrup) - Inner ear: cochlea and 3 semi-circular canals - sound waves are directed into the outer ear by the pinna & kinetic energy (set-up vibrations of identical frequency) in the tympanic membrane - this kinetic energy is transferred to the three small bones (hammer, anvil & stirrup) that amplify and transfer the energy into the oval window - this energy is then passed through the round window that relieves pressure in the cochlea - in the cochlea, hair cells bend when the soundwave moves through that transforms this wave into electrical energy that is sent to the auditory nerve to the brains → hearing loss - two main types - Conductive hearing loss - Factors that inhibit the movement of sound - Caused by malformation of structures in ear, infections in the outer/middle part of ear, ossicles trauma (sudden loud noises e.g. explosions or head trauma) - Sensorineural hearing loss - damage or malformation of the inner ear including cochlea, hair cells or auditory nerve - Usually permanent - Causes include excessive noise exposure, heredity, birth defects, infections, tumours, medication, aging - It prevents the kinetic energy being transformed into electric impulses sent to the brain for interpretation - technologies used to assist (next dotpoint)

- visual disorders



→ eye

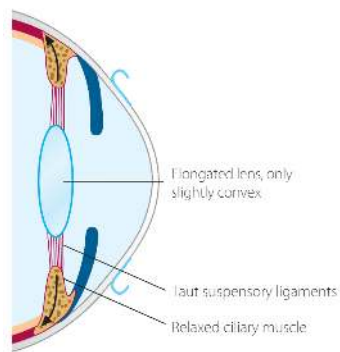
: receive light energy from the external environment and convert it to nerve impulses (electrical energy), which are sent via the optic nerve to the brain, where they are interpreted as images

Parts		Description
	Sclera	: forms the opaque protective layer (including front of eye: cornea and conjunctiva)
	Chloroid	: contains blood vessels, ciliary body, lens and iris
	Retina	: contains photoreceptors which detect light and convert light signals to electrochemical signals which can be interpreted by the brain
	Suspensory ligaments	: attach the ciliary muscles to the lens, allowing the lens to change shape
	Ciliary body	: controls thickness of lens as it controls the tension of suspensory ligaments
	Conjunctiva	: mucus membrane that covers front of eye to lubricate and avoid friction when blinking
	Iris	: controls the amount of light entering the eye by dilating or constricting pupil size
	Cornea	: curved casement covers front of eye that refracts $\frac{2}{3}$ light allowing focus
	Pupil	: allows light to go through to retina
	Lens	: transparent bi-convex lens that refracts light to achieve focus
	Aqueous humor	: nourishing eye through fluid
	Macula	: to process harp, clear, straight-ahead vision.
	Fovea	: greatest visual acuity (ability to see a clear and precise image)
	Optic nerve	: transmits electrical (visual) signals from the retina to the visual cortex in the brain
	Optic disk (aka. blind spot)	: optic nerves join with no phoresreceptors = brain fills missing parts to create whole image
	Vitreous chamber (including vitreous humor)	: clear fluid to refract light and maintain shape of eyeball

• accommodation

: In order to form a sharply focused image, on the retina, of objects at any distance, the lens must change its curvature to refract the light by the correct amount.

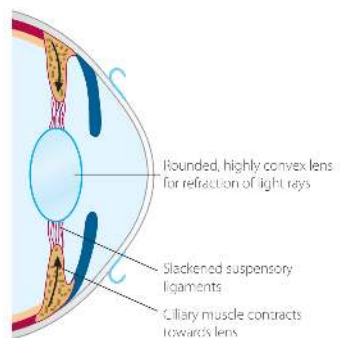
Diagram	Vision	Description
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Distant vision : the ciliary muscles relax, drawing the sclera back, which holds the suspensory ligaments taut

- this has the effect of lengthening the lens, decreasing the curvature

- reducing the refraction of the light.



Near vision : the curvature of the lens must be increased, to increase the refractive power of the lens

- achieved by the ciliary muscles contracting, which draws the sclera forward and releases the tension on the suspensory ligaments

- This allows the lens to become rounder, increasing the curvature

- refracting the light to a greater degree.

→ visual disorders

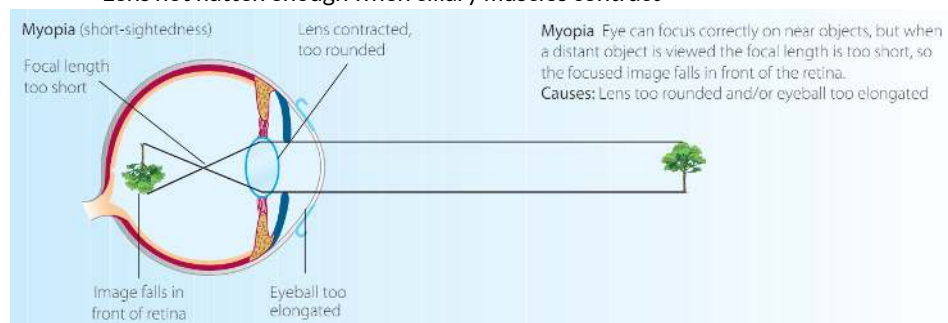
• myopia (short-sightedness)

: see objects near clearly but distant objects blurred

- the focused image falls in front of retina

- causes include

- Shape of eyeball is too elongated
- Inadequate refractive power of cornea
- Lens not flatten enough when ciliary muscles contract



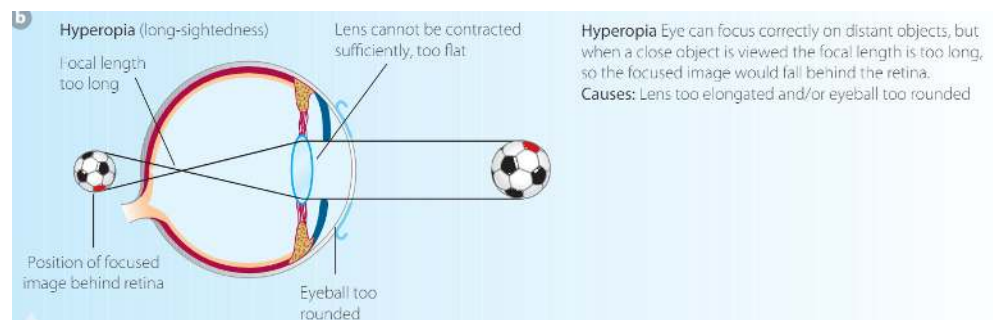
• hyperopia (long-sightedness)

: see distant objects but near objects are out of focus

- the focused image falls behind the retina

- causes include:

- Eyeball is rounded
- Lens is too flat
- Due to elasticity in old age
- Refractive power is too great



- cataracts

: clouding of the lens that reduces transmission of light through lens
 - blurred vision and increased sensitivity to sunlight

- macular degeneration (retinal pigment epithelium)

: a disease that causes degeneration of cells beneath retina
 - responsible for 50% of all causes of blindness & leading cause of legal blindness
 - the macula central vision is affected but peripheral vision is unaffected
 - prevents light from being focused successfully leading to loss of vision

- loss of kidney function

→ **kidneys**

: to remove nitrogenous wastes from the bloodstream and maintain optimal levels of substances such as salt and water (osmoregulation)

- functions (filtration, reabsorption & secretion)

- Excrete waste products
- Regulate salt and water balance
- Maintain acid balance
- Retain substances vital to the body (glucose, proteins, etc.)

- These are filtered out of the blood by nephrons within the kidney, which can be excreted from the body

- Excretion is essential to maintain other aspects of homeostasis, including maintaining the correct concentration of water and salts in the blood, blood volume and blood pressure

- Renal artery = brings blood into kidney

- Renal vein = brings blood out of kidney

- Ureter = takes urine out of kidney

Filtration	Reabsorption	Secretion	Excretion
: moving waste materials dissolved in the fluid of the blood from the blood vessels into the nephrons to be removed from the body (passive transport)	: essential components that were filtered out of the bloodstream are returned (amino acids, glucose, minerals, vitamins) via active transport.	: occurs in the distal tubules where substances actively from the blood into the nephron	: the water, nitrogenous wastes and other wastes that remain in the collecting duct are known as urine.
1. Blood flows into smaller blood vessels called capillaries 2. These capillaries form a clump called a glomerulus and is surrounded by the Bowman's capsule 3. Substances such as nitrogenous waste, water, glucose, salt and amino acids diffuse from the capillaries into the Bowman's capsule (start of the nephron)	• Proximal Tubule 1. Glucose and salt moves back into the capillaries via active transport 2. Drugs and hydrogen move into the nephron • Descending Loop of Henle 1. Water moves out of the nephron due to osmosis, creating a concentration gradient • Ascending Loop of Henle 1. Salt moves out of the nephron into the medulla due to diffusion, following	• Distal tubule 1. Toxic substances from the blood are actively moved into the tubules for removal e.g. urea, uric acid, ammonia, hydrogen ions, drugs 2. Filtrate then moves into collecting ducts which flow into the renal pelvis	- Urine collects in the renal pelvis - the urine travels from the pelvis through the ureters (tube) to the bladder, after which it is eliminated from the body through the urethra. - filtrate is now called urine.

	<table><tr><td>4. These substances are known as glomerular filtrate</td><td>the osmosis of water due to a concentration gradient (Urea and other wastes are not reabsorbed)</td><td></td><td></td></tr></table>	4. These substances are known as glomerular filtrate	the osmosis of water due to a concentration gradient (Urea and other wastes are not reabsorbed)		
4. These substances are known as glomerular filtrate	the osmosis of water due to a concentration gradient (Urea and other wastes are not reabsorbed)				
● investigate technologies that are used to assist with the effects of a disorder, including but not limited to: (ACSBL100) – hearing loss: cochlear implants, bone conduction implants, hearing aids	Hearing Loss: Technologies for Assistance ● Hearing Aids: : amplify sound to improve hearing for individuals with partial hearing loss. - components: Microphone, amplifier, speaker. - Mechanism: - Sound waves enter the microphone. - Sound is converted into electrical signals and amplified. - Amplified sound is transmitted into the ear through a speaker. - Limitations: - Only effective for people with functioning hair cells in the cochlea. - Not suitable for severe or profound hearing loss. - Applications: - Beneficial for individuals with mild to moderate sensorineural hearing loss. ● Cochlear Implants: : directly stimulates the auditory nerve using electrical signals. - components: External sound processor, internal implant, electrode array. - mechanism: - Sound processor captures sound and converts it into digital signals. - Signals are transmitted to the internal implant. - Electrodes in the cochlea stimulate the auditory nerve, bypassing damaged hair cells. - Benefits: - Suitable for individuals with severe to profound hearing loss who receive little benefit from hearing aids. - Limitations: - Requires surgery. - Results vary depending on age at implantation and duration of hearing loss. ● Bone Conduction Implants: : transmit sound vibrations directly to the cochlea via the skull bones. - components: External sound processor, internal implant (optional, depending on model). - mechanism: - External sound processor captures sound and converts it into vibrations. - Vibrations travel through the skull to the cochlea, bypassing the outer and middle ear. - Applications: - Suitable for individuals with conductive hearing loss or single-sided deafness. - Advantages: - Avoids issues related to middle ear problems. - Limitations: - Requires intact cochlear function. - Surgery may be required for implant models.				
– visual disorders: spectacles, laser surgery	Visual Disorders: Technologies for Assistance ● Spectacles (Glasses): : correct refractive errors by bending light to focus on the retina. - mechanism: - Lenses (concave for myopia, convex for hyperopia) adjust the path of incoming light. - Allows the image to be correctly focused on the retina. - Applications: - Used for conditions like myopia (short-sightedness), hyperopia (long-sightedness), astigmatism, and presbyopia. - Advantages: - Non-invasive and easily adjustable. - Widely accessible and cost-effective.				

- **Laser Eye Surgery (e.g. LASIK):**
 - : reshapes the cornea to correct refractive errors.
 - Mechanism:
 - A laser is used to remove a small portion of the corneal tissue, changing its shape.
 - This corrects how light is focused on the retina, addressing conditions like myopia, hyperopia, and astigmatism.
- Advantages:
 - Permanent vision correction in most cases.
 - Reduces or eliminates the need for glasses or contact lenses.
- Limitations:
 - Not suitable for everyone (e.g., patients with thin corneas or severe refractive errors).
 - Surgical risks such as infection, dry eyes, or visual disturbances.

- loss of kidney function:
dialysis

Loss of Kidney Function: Technologies for Assistance

● Dialysis:

: mimics kidney function by removing waste products and excess fluid from the blood when the kidneys fail to do so.

- types:

- Hemodialysis (figure a)
 - Blood is filtered outside the body using a machine.
 - Blood passes through a dialyzer (artificial kidney) which removes wastes and excess fluids.
 - Requires vascular access through a fistula or catheter.
- Peritoneal Dialysis (figure b)
 - Utilizes the body's peritoneal membrane as a filter.
 - A dialysis solution is introduced into the abdominal cavity through a catheter.
 - Waste products diffuse from the blood into the solution, which is then drained and replaced.

- Applications:

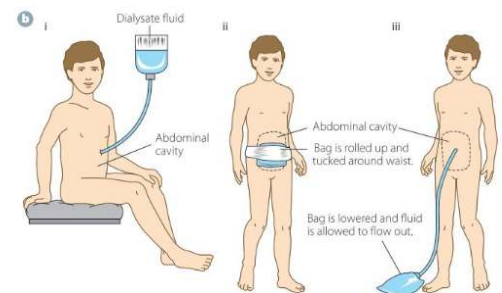
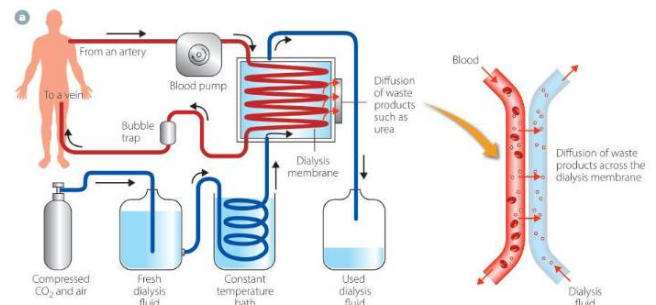
- Used in individuals with end-stage kidney failure or chronic kidney disease.

- Advantages:

- Life-saving for individuals waiting for a kidney transplant.

- Limitations:

- Time-consuming (hemodialysis usually takes 4 hours, 3 times a week).
- Requires strict dietary and fluid restrictions.
- Risk of infections, particularly in peritoneal dialysis.



- evaluate the effectiveness of a technology that is used to manage and assist with the effects of a disorder (ACSBL100)

Technologies

- some technologies may have side effects or require ongoing costs and maintenance, their benefits usually outweigh these downsides
- allow individuals with disorders to live more normal lives or extend their lifespan

→ cochlear implants

- Benefits:

- Enable deaf individuals to hear various sounds (e.g., alarms, cars), which are crucial for safety.
- Help children integrate into mainstream settings, allowing them to feel more comfortable.
- Support language acquisition and communication, especially effective for children.

- Limitations:

- Limited frequency range due to the small number of electrodes, so full hearing restoration is not possible.
- Require surgery, which carries risks such as infections or post-operative complications.

	- Transitioning to sound can be challenging, particularly for adults, where the effectiveness is reduced. → laser surgery - Benefits: - Quick procedure with fast recovery. - Corrects vision issues almost immediately. - In most cases (9/10), additional corrective eyewear is not necessary. - Limitations: - There's a small risk of complications (1/10), such as glare or dry eyes. - Individuals with strong lens prescriptions may not achieve complete vision correction. - Can be expensive, and there is a risk of damage or infection. → haemodialysis - Benefits: - Can be performed either in a hospital or at home. - Can be done while sleeping. Limitations: - Sessions last between 3-10 hours and must be done three times a week. - Requires the use of an external machine. - Potential side effects include low blood pressure, nausea, light-headedness, and more. → peritoneal dialysis - Benefits: - Provides uninterrupted time, allowing most patients to continue their daily activities, including working. - Can also be done while sleeping. - Limitations: - Requires the ability to set up and manage equipment independently. - Risks include infection at the catheter site, weight gain, hernias, weakening of abdominal muscles, and other complications.
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